

Function Power with Turtle

A. MULTIPLE CHOICE QUESTIONS

1. Observe the given program:

```
import math
a = int(input("Enter the number :"))
print("The square root is ", math.sqrt(a))
```

If the input is 25, what will be the output?

- a. 5
- b. 5.0
- c. 625
- d. 625.0

Answer: b. 5.0

2. Observe the given program:

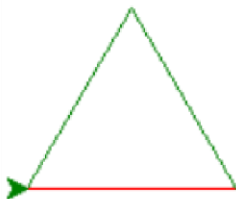
```
import random
print(???)
```

If the output is '0.9427568549552073', which of the following could be the function?

- a. random.random()
- b. random.randint()
- c. random.choice()
- d. random.shuffle()

Answer: a. random.random()

3. Observe the given program:



Which of the following could be the program?

- | | | | |
|----|---|----|---|
| a. | <pre>tom.color("red") tom.forward(100) tom.left(120) tom.forward(100) tom.left(120) tom.forward(100) tom.left(120)</pre> | b. | <pre>tom.color("green") tom.forward(100) tom.left(120) tom.forward(100) tom.left(120) tom.forward(100) tom.left(120)</pre> |
| c. | <pre>tom.color("red") tom.forward(100) tom.left(120) tom.color("green") tom.forward(100) tom.left(120) tom.forward(100) tom.left(120)</pre> | d. | <pre>tom.color("green") tom.forward(100) tom.left(120) tom.color("red") tom.forward(100) tom.left(120) tom.forward(100) tom.left(120)</pre> |

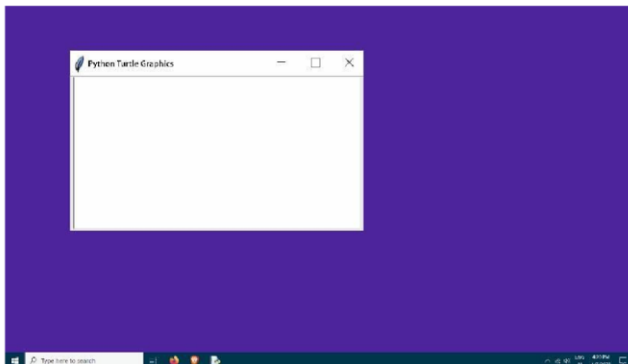
Answer: c.

4. Which of the following is a screen function in the turtle library?

- a. random()
- b. color()
- c. forward()
- d. bye()

Answer: d. bye()

5. Look at the following output:



Which of the following could be the correct arguments?

- a. s.setup(400, 600, 100, 100)
- b. s.setup(600, 400, 100, 100)
- c. s.setup(100, 100, 400, 600)
- d. s.setup(100, 100, 600, 400)

Answer: b. s.setup(600, 400, 100, 100)

B. FILL IN THE BLANKS

- 1. A library is a collection of modules that work together to solve complex problems.
- 2. The math module provides functions for complex mathematical operations.
- 3. The randint() function gives a random integer between the given integers.
- 4. The function which moves the turtle by a given number of steps in the direction it is facing is known as forward().
- 5. The screen functions can be used to change the background color, size, title, etc.

C. DESCRIPTIVE QUESTIONS

- 1. What are modules in Python? Explain with examples.**

A module is a set of functions related to do a specific task in Python. Some of the in-built modules are math() and random().

- 2. Explain the difference between random() and randint() functions, in the random module.**

The random() function is used to generate a random float number between 0 and 1. The number will be less than 1. On the other hand, the randint(a, b) function produces a random integer within the specified range , inclusive of a and b.

- 3. What is the turtle library? Explain the turtle functions : forward(), backward(), left() and right().**

The Turtle library is a Python library, used to create abstract patterns, logos, games, and science simulations with minimal code. It is a user-friendly, and captivating platform for learning and practicing Python programming.

The forward() function moves the turtle forward by a specified distance in the direction it is facing, the backward() moves it backward by a specified distance in the opposite direction it is facing, left() and right() rotate the turtle to the left(or right) by a specified angle.

4. Explain the various screen functions in the turtle module.

bgcolor() sets the background color of the screen according to the given argument, i.e. the name of the color like "red", "green", "blue", etc. setup() command sets the dimensions and position of the screen window. title() sets the title of the screen window. bye() command closes the screen automatically. exitonclick() command makes it so that the screen closes after we click it.

5. Write a turtle code to make a Rhombus.

```
t = turtle.Turtle()  
side=int(input("Enter side length:"))  
for i in range(2):  
    t.forward(side)  
    t.left(120)  
    t.forward(side)  
    t.left(60)
```

D. HOTS (HIGHER ORDER THINKING SKILLS) QUESTIONS**1. Write a code to make 5 squares of side 50 each. Also, use different colors for all 5 squares.**

```
import turtle  
t = turtle.Turtle()  
colors = ["red", "green", "blue", "orange", "purple"]  
for color in colors:  
    t.color(color)  
    for i in range(4):  
        t.forward(50)  
        t.left(90)  
    t.penup()  
    t.forward(60)  
    t.pendown()
```

2. Look at this turtle code, and draw its output.

```
import turtle
tom = turtle.Turtle()
tom.forward(100)
tom.left(120)
tom.forward(100)
tom.left(120)
tom.forward(100)
tom.left(120)
tom.penup()
tom.goto(150,150)
tom.forward(100)
tom.left(120)
tom.forward(100)
tom.left(120)
tom.forward(100)
tom.left(120)
```

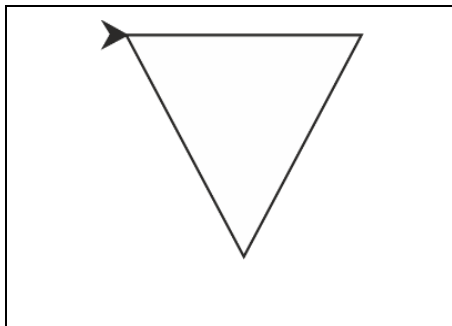
Output:



E. IMAGE-BASED QUESTIONS

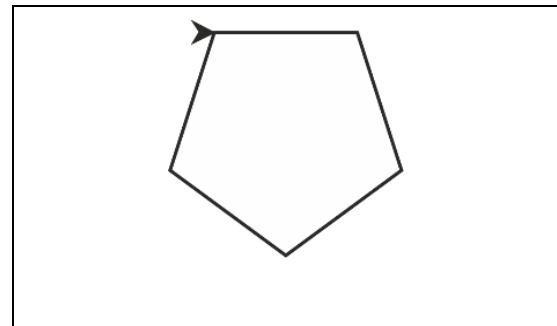
1. Examine the following outputs, and write the code:

a.



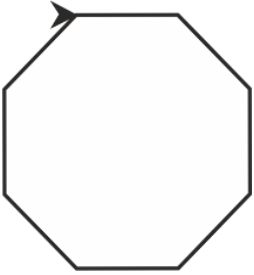
```
import turtle
tom = turtle.Turtle()
for i in range (3):
tom.forward(100)
tom.right (120)
```

b.

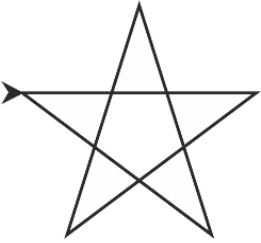


```
import turtle
tom = turtle.Turtle()
for i in range (5):
tom.forward(100)
tom.right (72)
```

c.

	<pre><u>import turtle</u> <u>turtle</u> for i in range (8): <u>tom.forward(100)</u> <u>tom.right (45)</u></pre>
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d.

	<pre><u>import turtle</u> <u>turtle</u> for i in range (5): <u>tom.forward(100)</u> <u>tom.right (144)</u></pre>
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F. UPSKILL

1. Explore 3 Python Libraries and try to make some interesting programs out of it.